Evaluation of 1D-CNN Architectures for Imbalanced Underwater Acoustic Motor-State Recognition

***Abstract*—Underwater acoustic data faces severe scarcity due to it's ever changing environment and when an already imbalanced dataset is divided into multiple motor specific tasks, it faces more severity of imbalance. Accuracy alone can not define the model's performance in this case. To evaluate whether the model can genuinely recognize the minority classes or not, macro f1 score is needed. This work approaches some architectures of 1D CNN model including ablation study through macro f1 score to compare those architectures and observe their effect on macro-f1. To improve minority-class detection, different architectural modifications are investigated in a shared 1D-CNN framework and then dedicated 1D CNN framework is evaluated for individual motors. Underwater motor acoustic data from different kind of motors, engine and propellers are used to analyze the performance of each design. Overall, the result shows that appropriate architectural modification using same 1D-CNN works better than a shared architecture based on the consistency of macro-F1 score performance.**

***Index Terms*—1D CNN, underwater acoustic target recognition, multi-task, imbalanced data, minority-class recognition, macro F1 score.**

I. Introduction

Underwater acoustic target recognition(UATR) detects the category or features of the underwater acoustic signal. UATR is now used in the marine search and rescue, seabed mapping and other marine applications too, crossing the boundary of military applications [1]. As per the recent perspectives, UATR is a very important research area for ocean remote sensing where the type and information of the targets can be achieved by analyzing underwater acoustic signal [2].

The biggest problem for underwater acoustic recognition is the limitation of the availability of authentic underwater acoustic recordings. It is really hard to collect the diverse and properly labeled recording from the underwater environment [3]. Also the data acquisition, storage and labeling problem can make the training data insufficient for the dataset. Learning different kind of acoustic patterns becomes difficult for the data-driven model when the authentic underwater recordings are limited. Therefore, practical robustness or generalization of the model gets affected. Machine Learning methods can recognize targets between the raw acoustic signal and label by learning the mapping. Statistical learning, deep learning, transfer learning and data augmentation are used in UATR research. Machine learning and deep learning based AI approaches are investigated in UATR due to the limitations of traditional signal-processing approaches. In deep learning based UATR research, CNN is used as a significant model family. CNN can learn the discriminative or local patterns automatically that lies in the acoustic signal. Deep learning model can work with two common input form for UATR: Time-domain signal and Time-frequency representation, which proves the relationship of model architecture with input representation [4].

1D-CNN can learn the temporal patterns by imposing convolution on the sequential or time-domain acoustic signal. It means without converting the signal into spectrogram, temporal acoustic information can be modeled. Time-domain-based models can have some practical advantages such as: smaller model/storage requirements, faster inference and suitability of resource-constrained deployment. But it doesn't claim that 1D-CNN is better than 2D-CNN, rather time-frequency representation can give higher accuracy in some cases, though computational or storage cost can be higher [4]. A recent study in 2025 has proposed a lightweight 1D-CNN with attention mechanism that used multi-scale residual convolution, attention mechanism and feature fusion with 1D CNN to improve the feature representation [5]. This proves researchers are also investigating how the CNN architecture should be designed.

In traditional single-task learning, models generally learn the features for a primary task. On the other hand, the same model can do multiple related tasks simultaneously in multi-task learning. Here, some features can be shared, and some can be task specific also. Useful information from one task can help the representation learning of another task, which is the motivation of multi-task learning [6]. A study in 2023 has directly used multi-task learning while proposing robust feature extraction for UATR using real-environment dataset and the robustness of the method was examined in different noise level [6]. Another study in 2024 proposed multi-task, multi-gate and multi-expert framework while trying to train the model robust representation on shared parameters between auxiliary and recognition task, which made a scope to use different parameter spaces for underwater signal. This says complex architecture can improve representation.

Though the previous researches have shown the effectiveness of multi-task learning or several sophisticated architectures, the scope for the systematic comparative evaluation of multi-task 1D-CNN architectural designs in the same experimental framework for underwater motor-state recognition still remains [3]- [6].

Only accuracy can be misleading in the imbalanced classification. Because high accuracy means the model predicts

majority classes more accurately. Specifically, deep learning models are not even fully free from this issue. The performance of the minority or tailed classes can be low [7]. Deep learning can't prevent the class-imbalance problem automatically, despite being succeeded for many times. The minority classes can still be unrecognized [8]. Macro F1 is very essential metric for performance evaluation in this case, because it gives same importance to every class, since it has the average F1 score of each class [9]. That's why this study considers macro F1 score while comparing multi-task 1D-CNN architectures for underwater acoustic motor state recognition.

Data level oversampling is the elementary or the widely accepted solution for the imbalanced audio classification, which was established by Chawla and the other co-authors by their SMOTE method, where new synthetic samples were created by using k-nearest-neighbor in between the existing minority class sample. This prevents model overfitting [10]. Implementing both random oversampling and SMOTE on the extracted embedding vectors instead of raw audios proves feature level oversampling is equally beneficial to audio tasks [11]. The implementation of FDLP-based feature extraction with SMOTE gives good result in the audio domain, which proves domain-independent efficiency of this method [12]. But when the number of features are too low, adding noise or making the synthetic samples systematically like SMOTE is much better than just making duplicate samples as it is [13]. Synthetic samples are created in between the triangular area by using cluster-center and two neighbors in the next study for this reason [14]. Methods like cosine-weighted cross entropy [15] and focal loss [16] has performed better in ShipsEar for the loss-level solution, resampling-based TResNet pipeline [17] of UAMTR, the sample-weight-encoding classifier [18] of PBCC, and intra-class and inter-class diversified expert-routing [19] of Mixture-of-Experts has given notable result for the architecture-level solution. Since the general method has performed insufficiently in the extreme imbalance, DQNiDTC has reconstructed the classification as reinforcement learning problem [20], and inverse-square-root weighting in long-tailed data [21], custom CIoU-DFL loss [23] in sonar detection and similarity-based sampling is used in fisheries acoustic data.

This work creates and evaluates several architectural design for multi-task 1D-CNN of underwater motor-state recognition. Besides the overall classification performance, macro f1 score is used for analyzing class-wise balanced performance. In this way, which multi-task architecture can give balanced recognition between minority and majority motor states is analyzed here.

II. System Architecture

Several configuration is used here such as configuration with data augmentation with noise jittering, loss function testing, implementing the dense layer, batch normalization, dropout in several ways in the 1D CNN model. But before that data acquisition, pre-processing, visualization is necessary.

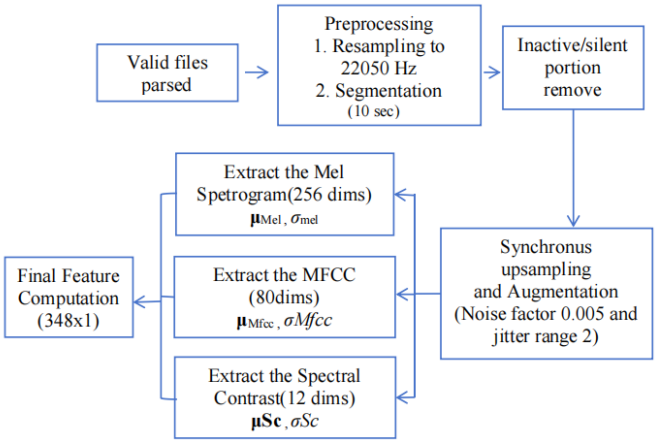


Fig. 1. Data processing flow chart.

A. Data Loading

In this experiment Wolfset: A High-Quality Underwater Acoustic Dataset for Algorithm Development and Analysis [23] was chosen for it's real underwater acoustic recordings of multiple propulsion-related sound sources, such as outboard motors and an electric motor associated with small-vessel environments collected in controlled laboratory conditions incorporating realistic noise and transient disturbance sources. It has 12 multi-source and 24 distinct classes of motor sounds. The sample distribution of these classes are very imbalanced such as some classes include only 1 sample against 46 samples of one class. Motor 1 to 4 out of 5, has the operating conditions with the annotated value 0 to 5 representing Absent, Idle disengaged, Idle engaged forward, medium forward and varying and motor 5 has the values from 0 to 4 which states the Absent, Slow forward, Fast Forward, Slow reverse and Fast reverse sequentially. In this study, the multi-task classification focuses on recognizing the 5 motors with their operating conditions.

B. Data Preprocessing

In this part, all the audios were resampled to 22050 Hz. Files where only the background exist were filtered out to get balance. These 46 files with the rest of 122 making the large number of background majority classes. This didn't create any issue, because we are getting the background classes also from the 122 files along with the motor sound. After that, each of the 122 audio files was segmented into 10 seconds. The silent or inactive portion from each segment was removed. This data goes to the extraction part for the ablation study which is configuration A. For the configuration B,C and D, synchronus oversampling[rare classes increased toward target count] and a subtle physics aware augmentation with gaussian noise(0.005) and time shift jitter(-2 to +2) is done on each of the segments. This creates more samples and the training process closer to realistic underwater conditions.

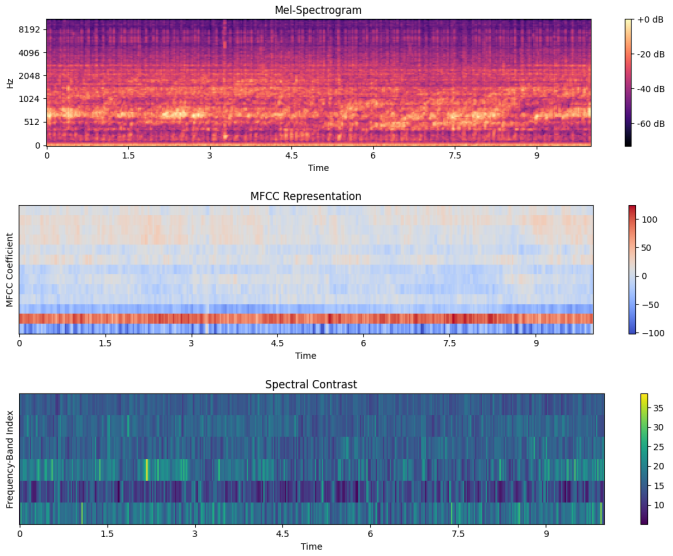


Fig. 2. Mel-spectrogram, MFCC, and spectral contrast representation of a segment of motor 2(state:idle engaged, transient: shot, Observation: 4th shot failed).

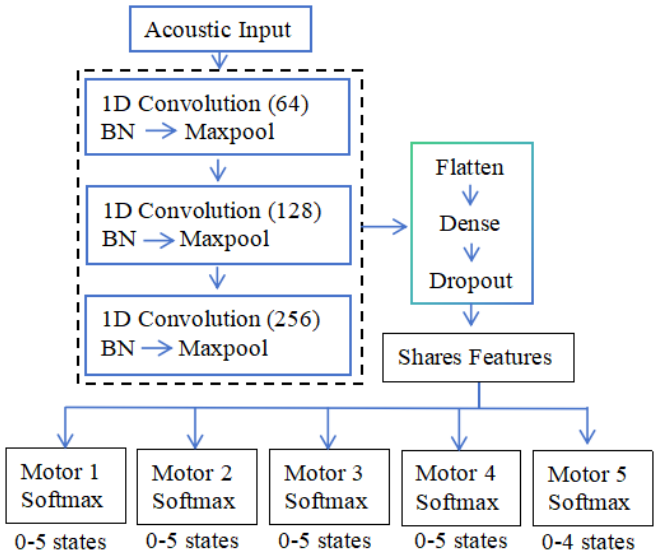


Fig. 3. General architecture of Shared Multi-Task Backbone Network Pipeline.

C. Feature Extraction

Parameters for the feature extraction in the implementation were: 2048 fft length,4096 hop length, 512 mel bands, 128 mel frequency cepstral coefficients and 40 spectral contrast bands. It then calculated the summarized mean and standard deviation of 348 feature groups over time, resulting in an extracted matrix consisted of 1,665 segmented samples and a single 348 × 1 feature vector for every audio segment spanning ten seconds. This structure gave the model ample opportunity to learn unique average spectral behavior and temporal variability for each underwater acoustic segment. The whole data processing and feature extraction flow is shown in Fig. 1. The feature representation is shown in Fig. 2 for an example.

D. Network Construction

Due to the sequential structure of the feature vector and lightweight architecture, a 1D convolution model is chosen for this work which will learn local relationships among neighboring acoustic descriptors. The 1D-CNN performs convolutions over the input array to pull out shift-invariant acoustic features through shifting local learning filters. To allow the network to model non-linear acoustic properties, a Rectified Linear Unit (ReLU) activation function is applied across all intermediate processing layers. For the final output layers, a Softmax activation function normalizes raw predictions into a valid probability distribution. Network optimization is driven by the Adam (Adaptive Moment Estimation) algorithm. Adam computes adaptive learning rates for each parameter by tracking exponentially decaying averages of past gradients and squared gradients [24].

E. 1D-CNN Configurations

The 1D-CNN model is applied in some phases with some changes in it's design for evlauating the macro F1 score.

- **Configuration A**: This is the generalized configuration with the baseline loss sparse categorical cross-entropy for the comprehensive classification that is shown in Fig. 3. As per the ablation study focal loss and weighted loss is applied to observe the macro f1 score. Focal loss is tried because it down weights the relative impact of easy examples and puts more focus on difficult or misclassified samples [25]. Weighted loss was evaluated to penalize errors in the classes for which fewer examples were available.
- **Configuration B**: Here synchronous up-sampling and a subtle physics aware data augmentation is done with gaussian noise 0.005 and time shift jitter in the range of 2(-2 to +2) and for training, the same generalized model configuration shown in Fig. 3 is used with the loss function that worked well in the ablation study. This synchronous resampling and Gaussian jitter is imposed to reduce the imbalance in the training data at the folds but the validation data are kept unchanged. This augmentation and up-sampling method is also applied to configuration C and D.
- **Configuration C (Shared Backbone with Task-Specific Crucial Deep Towers)**: In the architecture, shown in Fig. 4 a shared acoustic feature representation is created for all the motors by using three convolution 1D blocks. After that, individual task specific tower is created for each motor from this shared representation where each tower includes Dense(128), Batch normalization and Dropout(0.4) to classify the states for corresponding motors. In this way, each motor can do classification according to it's own feature by using the same shared feature.
- **Configuration D (Individual CNN)**: This architecture shown in Fig. 5, is not just implemented to increase the

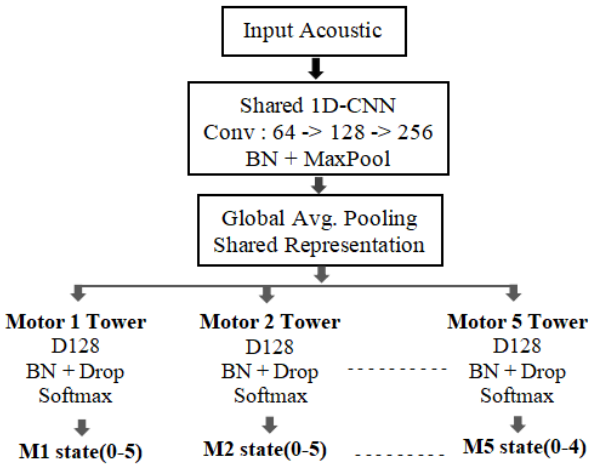


Fig. 4. Configuration C: Shared backbone with Task-Specific Crucial Deep Towers.

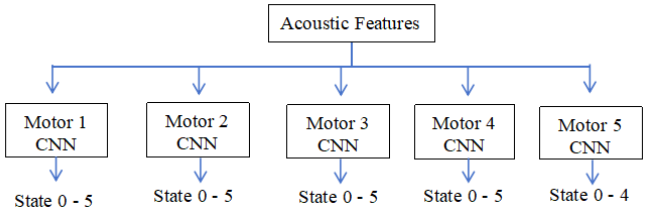


Fig. 5. Configuration D: individual CNN for Multi-Task Network Pipeline.

TABLE I
COMMON TRAINING SETTINGS.

Parameter	Value
Optimizer	Adam
Learning rate	0.001
Batch size	32
Epochs	100
Activation	ReLU
Output	Softmax

number of parameters in the model. Instead, the purpose is to adapt this model for the task structure of Wolfset. In this configuration, individual 1D-CNN is created and trained for each motor, where acoustic features are 1st processed by two convolution block for feature extraction. After that, motor state is classified for corresponding motor by using dense layer and dropout.

To prevent data leakage, the acoustic segments of the same recording sessions are kept either only in the training set or in the validation set. This was done for all the configurations. Thus tells the performance of the model on the unseen recording sessions. Table I shows all the common hyper meters used through all the configurations.

III. RESULT AND DISCUSSION

In the configuration A, as per the ablation study three types of loss functions are tested to see which works better. For motor 3,4 and 5 focal loss worked better and for 1 and 2 standard cross entropy worked better as we can see in Fig.6.

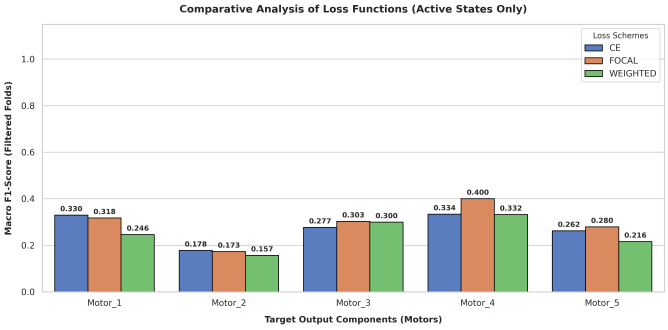


Fig. 6. Macro-F1 comparison of loss functions across motor-specific head at the baseline configuration A.

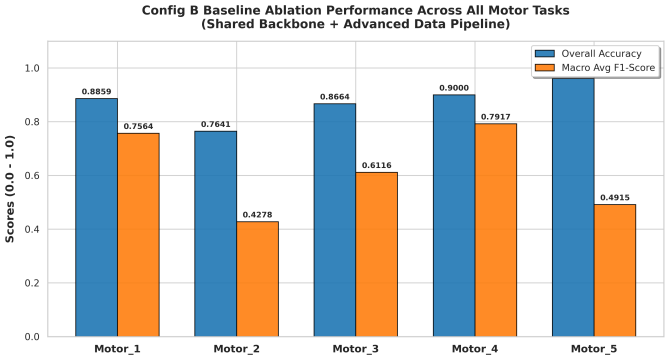


Fig. 7. Accuracy and macro-F1 comparison of configuration B.

Motor 1, 3 and 4 shows notable improvement but motor 2 and 5 however stayed weak, in the configuration B where synchronous up-sampling and physics aware augmentation is used. The result is shown in Fig.7. Configuration C, which is task specific towers, performed much better than the previous two configurations for motor 1,3 and 5. It also improved motor 2 slightly as per the result shown in c. Motor 4 however decreased relative to the physics-augmented configuration.

It shows that while task-specific representation is valuable, it does not straightforwardly resolve all motor head issues. In addition, the tower helped motor states that had specialized decision boundaries, but some performance was still dependent on data distribution and acoustic separability. The fully shared model was overly simplistic; the task-specific extension provided more flexibility to the model. However, the differing performance of motors still hinted that deeper towers could not fully solve shared-representation shortcomings. So the motivation for switching over to Configuration D is obvious from the earlier experiments. The performance of this architecture is shown in Fig.9. Macro f1 gives overall best performance to this configuration. The macro f1 score of each state of the motor is significantly greater than the shared backbone and share tower configuration which is shown in Fig.10. Table II shows how this architecture identifies each motor states through f1-score.

Some minority states are limited to one recording session. That's why, in the session-level validation split they are not present, which makes the zero support for the states and

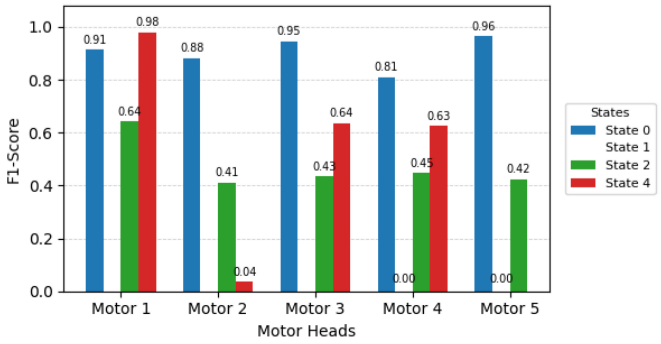


Fig. 8. Pooled classification report (support > 0) of macro-F1 score of each motor state of the task specific tower(configuration C).

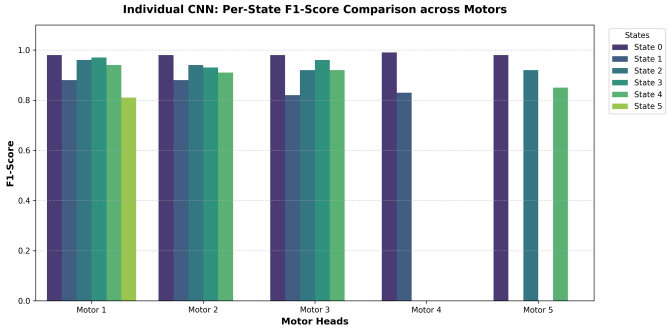


Fig. 9. Per state macro f1 score of configuration D (support > 0).

eventually are not represented in the classification report. In the configuration A and B 5 motors share same samples and same session-level split. On top of that the split is stratified based on synchronous multi-motor label combination. That's why, the state 1 of motor 1 and 3 is not present in the validation set, though they existed in the training data. In the dedicated CNN(Configuration D), each motor is treated as separate dataset. That's why, if the session containing state 1 of a particular motor is assigned to a validation fold, state 1 of that motor will appear in the evaluation.

Our result indicates that a motor-specific acoustic modeling is better suited for this dataset than having a heavily shared multi-task representation. Each motor seem to have own acoustic signature and class-distribution properties. Having a dedicated stream for each motor would enable the model to learn features discriminative-only to that motor head and not be limited in terms of output space by another motor head. But the Configuration D also comes at a price. It took the most parameters. Thus, it has a higher computational cost than Configuration A, B and C. So from the perspective of pure performance Configuration D is strongest model. The lighter configurations might still be useful from a deployment perspective when memory, latency, or hardware limitations are a concern. Overall, Configuration D dominated the standings as it yielded the most replicable motorwise Macro F1 performance and was just simply methodologically honest. It was not reliant on a misleading gain in accuracy rather it improved all five motor heads consistently.

TABLE II
F1-SCORES OF INDIVIDUAL MOTOR-STATE CLASSIFIERS ACROSS ALL THE STATES (SUPPORT > 0).

Motor	State 0	State 1	State 2	State 3	State 4	State 5	Macro F1
Motor 1	0.98	0.88	0.96	0.97	0.94	0.81	0.923
Motor 2	0.98	0.88	0.94	0.93	0.91	–	0.928
Motor 3	0.98	0.82	0.92	0.96	0.92	–	0.920
Motor 4	0.99	0.83	–	–	–	–	0.910
Motor 5	0.98	0.00	0.92	–	0.85	–	0.917

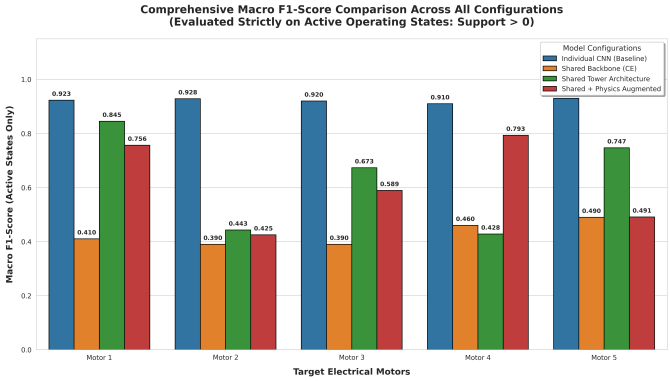


Fig. 10. Comprehensive macro f1 comparison across all configurations(evaluated strictly on active operating states: support > 0.

IV. Conclusion

This study presents a phase by phase evaluation of the deep learning based 1D-CNN frameworks for multi-task underwater motor-state recognition in the severe data imbalance. The initial phase includes a thorough data cleaning and preprocessing of the exact multi-task extraction from the single combined distribution of the Wolfset dataset. The model predicts the operating state of underwater motors of small vessels instead of predicting a single combined label. This multi-task prediction made the task more complex but interpretable and better aligned with dataset structure. The experimental results through structured progression indicate that the overall motor-specific approach outperformed entirely shared learning in this case since we have five types of motors. While the shared backbone of a single CNN motor-wise setup yielded weak results through Macro f1-score values. This demonstrates that every motor has unique acoustic feature and benefit from learning the representation by an individual motor.

V. Limitations

The initial study relied exclusively on the Wolfset dataset which further needs to be validated on other independent underwater acoustic datasets before generalization claims. Secondly, the dataset was made in a controlled underwater setting. Actual open-water conditions could have an impact on model performance. Due to the imbalance some samples were from only one session so at a time either the training set or the validation set got that session which made that state of zero support and eventually out of the pooled classification report. The fourth one is the last high performance model uses more parameters than the shared configurations. However

since we got only 5 types of motors here so this didn't create any computational issue. But this results in a trade-off between performance and computation cost when there are more types. Lighter models might also be tested for real-time or low-power deployment.

VI. Future Directions

In future this work can be extended along several avenues. 1st of all this framework should be evaluated in different datasets in underwater acoustic. Secondly, if more rare motor-state samples can be collected or generated through carefully controlled acoustic augmentation, a hierarchical multi task motor state recognition can be done. It will work like this: 1st of all it will tell which motor is on and then it's operating condition. The recognition will be done in two stages, if the motor prediction is true, only then it can predict true states. And finally, explainable AI methods can be applied to know which frequency bands or feature groups are most influential in each motor-state prediction.

References

[1] X. Luo, L. Chen, H. Zhou, and H. Cao, "A Survey of Underwater Acoustic Target Recognition Methods Based on Machine Learning," JMSE, vol. 11, no. 2, p. 384, Feb. 2023, doi: 10.3390/jmse11020384.

[2] S. Feng, S. Ma, X. Zhu, and M. Yan, "Artificial Intelligence-Based Underwater Acoustic Target Recognition: A Survey," Remote Sensing, vol. 16, no. 17, p. 3333, Sep. 2024, doi: 10.3390/rs16173333.

[3] Y. Xie, J. Ren, J. Li, and J. Xu, "Advancing robust underwater acoustic target recognition through multitask learning and multi-gate mixture of experts," The Journal of the Acoustical Society of America, vol. 156, no. 1, pp. 244–255, Jul. 2024, doi: 10.1121/10.0026481.

[4] J. Tang, W. Gao, E. Ma, X. Sun, and J. Ma, "Deep Learning Based Underwater Acoustic Target Recognition: Introduce a Recent Temporal 2D Modeling Method," Sensors, vol. 24, no. 5, p. 1633, Mar. 2024, doi: 10.3390/s24051633.

[5] ZHANG Yufei, "Underwater acoustic target recognition based on one-dimensional attention mechanism of convolutional neural network," Journal of Underwater Unmanned Systems, vol. 33, no. 5, p. 905, 2025, doi: 10.11993/j.issn.2096-3920.2025-0053.

[6] D. Li, F. Liu, T. Shen, L. Chen, and D. Zhao, "A Robust Feature Extraction Method for Underwater Acoustic Target Recognition Based on Multi-Task Learning," Electronics, vol. 12, no. 7, p. 1708, Apr. 2023, doi: 10.3390/electronics12071708.

[7] W. Chen, K. Yang, Z. Yu, Y. Shi, and C. L. P. Chen, "A survey on imbalanced learning: latest research, applications and future directions," Artif Intell Rev, vol. 57, no. 6, p. 137, May 2024, doi: 10.1007/s10462-024-10759-6.

[8] K. Ghosh, C. Bellinger, R. Corizzo, P. Branco, B. Krawczyk, and N. Japkowicz, "The class imbalance problem in deep learning," Mach Learn, vol. 113, no. 7, pp. 4845–4901, Jul. 2024, doi: 10.1007/s10994-022-06268-8.

[9] A. A. Khan, O. Chaudhari, and R. Chandra, "A review of ensemble learning and data augmentation models for class imbalanced problems: Combination, implementation and evaluation," Expert Systems with Applications, vol. 244, p. 122778, Jun. 2024, doi: 10.1016/j.eswa.2023.122778.

[10] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," jair, vol. 16, pp. 321–357, Jun. 2002, doi: 10.1613/jair.953.

[11] D. Liang and E. Thomaz, "Audio-Based Activities of Daily Living (ADL) Recognition with Large-Scale Acoustic Embeddings from Online Videos," Proc. ACM Interact. Mob. Wearable Ubiquitous Technol., vol. 3, no. 1, pp. 1–18, Mar. 2019, doi: 10.1145/3314404.

[12] M. Dua, S. Joshi, and S. Dua, "Data augmentation based novel approach to automatic speaker verification system," e-Prime - Advances in Electrical Engineering, Electronics and Energy, vol. 6, p. 100346, Dec. 2023, doi: 10.1016/j.prime.2023.100346.

[13] Z. Luo, X. Zeng, Z. Bao, and M. Xu, "Deep Learning-Based Strategy For Macromolecules Classification with Imbalanced Data from Cellular Electron Cryotomography," in 2019 International Joint Conference on Neural Networks (IJCNN), Budapest, Hungary: IEEE, Jul. 2019, pp. 1–8. doi: 10.1109/IJCNN.2019.8851972.

[14] J. Lyu, J. Yang, Z. Su, and Z. Zhu, "LD-SMOTE: A Novel Local Density Estimation-Based Oversampling Method for Imbalanced Datasets," Symmetry, vol. 17, no. 2, p. 160, Jan. 2025, doi: 10.3390/sym17020160.

[15] Y. Ma et al., "Imbalanced Underwater Acoustic Target Recognition with Trigonometric Loss and Attention Mechanism Convolutional Network," Remote Sensing, vol. 14, no. 16, p. 4103, Aug. 2022, doi: 10.3390/rs14164103.

[16] Z. Chen, G. Xie, M. Chen, and H. Qiu, "Model for Underwater Acoustic Target Recognition with Attention Mechanism Based on Residual Concatenate," JMSE, vol. 12, no. 1, p. 24, Dec. 2023, doi: 10.3390/jmse12010024.

[17] L. Chen, X. Luo, H. Zhou, Q. Shen, L. Chen, and C. Huan, "A robust deep learning model for underwater acoustic multi-target recognition under non-target interference conditions," Expert Systems with Applications, vol. 296, p. 129122, Jan. 2026, doi: 10.1016/j.eswa.2025.129122.

[18] Q. Yin and L. Shen, "Underwater acoustic target recognition based on population balance-encoding classification," Ocean Engineering, vol. 337, p. 121899, Oct. 2025, doi: 10.1016/j.oceaneng.2025.121899.

[19] Y. Xie, J. Ren, and J. Xu, "Unraveling complex data diversity in underwater acoustic target recognition through convolution-based mixture of experts," Expert Systems with Applications, vol. 249, p. 123431, Sep. 2024, doi: 10.1016/j.eswa.2024.123431.

[20] Z. Liu, C. Hu, B. Xu, M. Huang, and Z. Xiao, "Underwater acoustic target detection with imbalanced dataset based on deep Q-network with dual termination criterion," Engineering Applications of Artificial Intelligence, vol. 160, p. 111661, Nov. 2025, doi: 10.1016/j.engappai.2025.111661.

[21] J. Yu, S. Yan, L. Mao, Z. Sui, and C. Jiang, "Modulation feature enhancement with a multi-stage attention network for underwater acoustic target recognition," Signal Processing, vol. 247, p. 110669, Oct. 2026, doi: 10.1016/j.sigpro.2026.110669.

[22] Z. Wang, J. Guo, S. Zhang, and Y. Zhang, "Sonar-based object detection for autonomous underwater vehicles in marine environments," Front. Mar. Sci., vol. 12, p. 1539371, Apr. 2025, doi: 10.3389/fmars.2025.1539371.

[23] V. Lobo, N. Pessanha Santos, and R. Moura, "A High-Quality Underwater Acoustic Dataset for Algorithm Development and Analysis," Sci Data, vol. 12, no. 1, p. 1323, Jul. 2025, doi: 10.1038/s41597-025-05564-x.

[24] C. Bellinger, R. Corizzo, and N. Japkowicz, "Performance Estimation bias in Class Imbalance with Minority Subconcepts," in Proceedings of the Fifth International Workshop on Learning with Imbalanced Domains: Theory and Applications, PMLR, Jun. 2024, pp. 31–44. Accessed: Aug. 27, 2026. [Online]. Available: https://proceedings.mlr.press/v241/bellinger24a.html

[25] J. Hildebrand, "Anthropogenic and natural sources of ambient noise in the ocean," Mar. Ecol. Prog. Ser., vol. 395, pp. 5–20, Dec. 2009, doi: 10.3354/meps08353.